Saturation of the Right-Leg Drive Amplifier in Low-Voltage ECG Monitors (April 2014)

Daniel K. Freeman*, Ronald D. Gatzke, Georgios Mallas, Yu Chen, and Chris J. Brouse

***Abstract*—Electrocardiogram (ECG) monitoring is a critical tool in patient care, but its utility is often balanced with frustration from clinicians who are constantly distracted by false alarms. This has motivated the need to readdress the major factors that contribute to ECG noise with the goal of reducing false alarms. In this study, we describe a previously unreported phenomenon in which ECG noise can result from an unintended interaction between two systems: 1) the dc lead-off circuitry that is used to detect whether electrodes fall off the patient; and 2) the right-leg drive (RLD) system that is responsible for reducing ac common-mode noise that couples into the body. Using a circuit model to study this interaction, we found that in the presence of a dc lead-off system, even moderate increases in the right-leg skin–electrode resistance can cause the RLD amplifier to saturate. Such saturation can produce ECG noise because the RLD amplifier will no longer be capable of attenuating ac common-mode noise on the body. RLD saturation is particularly a problem for modern ECG monitors that use low-voltage supply levels. For example, for a 12-lead ECG and a 2 V power supply, saturation will occur when the right-leg electrode resistance reaches only 2 MΩ. We discuss several design solutions that can be used in low-voltage monitors to avoid RLD saturation.**

***Index Terms*—Electrocardiogram (ECG), electrode impedance, lead-off, right-leg drive (RLD).**

I. Introduction

ELECTROCARDIOGRAM (ECG) monitors are ubiquitous in modern clinical environments, and their ability to convey reliable, low-noise signals is fundamental to patient monitoring. Because electrical noise can couple into the ECG and interfere with clinical monitoring, there has been significant effort over many decades to understand the origin of such noise [1]. This issue has been given renewed attention due to the fact that ECG noise can trigger false alarms, contributing to the growing problem of "alarm fatigue" [2]. It has been estimated that over 80% of all alarms from patient monitors are false positive, and these excessive alarms can cause clinicians to be distracted from their work and become desensitized to alarms in general, ultimately increasing the number of treatment errors [2], [3], [17]. We hypothesized that ECG monitors being designed today have to contend with a source of ECG noise that was not present even a decade ago. Specifically, there have been recent changes in the semiconductor industry in which the voltage supply levels used to power integrated circuits are much lower than in years past. These lower supply voltages could potentially cause amplifiers in the ECG front-end circuitry to saturate, producing noise on the ECG. To test this hypothesis, we built a circuit model of an analog ECG front-end. We found that, indeed, such saturation can occur as a result of the unintended interaction between two separate circuits: 1) the right-leg drive (RLD) circuitry that suppresses ac common-mode noise on the body [4], [5]; and 2) the dc lead-off system that detects when any electrodes have fallen off of the patient.

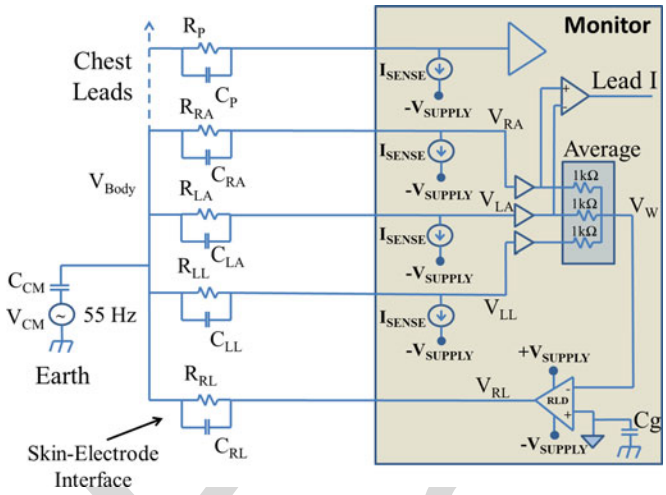


Fig. 1. Illustration of the circuit model that we used to represent a typical ECG system. The body is coupled capacitively to the power mains (V_{CM}) through capacitor C_{CM}. The skin–electrode interface of each electrode is represented by a parallel resistor and capacitor. The RLD system works by averaging the voltages from the primary electrodes (V_{RA}, V_{LA}, V_{LL}) to produce the Wilson voltage (V_W), which is then applied to the inverting input of the RLD amplifier. The dc lead-off system works by pulling current (I_{sense}) through each lead and monitoring the dc voltage on each electrode. The precordial leads are present for the ten-electrode tests but not the four-electrode tests.

To understand how this interaction occurs, it is necessary to first understand how these two systems work in isolation. The RLD system was first introduced into commercial monitors in the 1960s, and it was designed to address the most prominent ECG noise source, that of the mains power (50 Hz in Europe, 60 Hz in the United States). This noise couples into the body through parasitic capacitance to the mains, producing a large ac voltage on the body that is common to all electrodes, hence serving as a source of common-mode noise in the ECG (see Fig. 1). The RLD system works by taking the average voltage appearing on the primary electrodes (right-arm (RA), left-arm (LA),

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*D. K. Freeman is with the Department of Algorithms, Dräger Medical Systems, Inc., Andover, MA 01810 USA (e-mail: danielkfreeman@gmail.com).

R. D. Gatzke is with Gatzke Technologies, Lexington, MA 02421 USA (e-mail: ron@gatzketech.com).

G. Mallas, Y. Chen, and C. J. Brouse are with the Department of Algorithms, Dräger Medical Systems Inc., Andover, MA 01810 USA (e-mail: Georgios.Mallas@draeger.com; Yu.Chen@draeger.com; Chris.Brouse@draeger.com).

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left-leg (LL)), inverting this signal, and then applying this voltage to the right-leg (RL) electrode to effectively cancel out the ac common-mode noise source on the body [4], [5]. The amplifier that is responsible for driving the inverted signal back into the RL electrode is referred to as the RLD amplifier.

Designers of ECG front-end circuitry must ensure that the RLD amplifier does not saturate on its output during normal operation, or else the RLD system will not function properly. Up until recently, this has not been a major problem because the entire ECG front-end circuitry was typically powered with high voltages (e.g., ±5–10 V), and therefore the RLD amplifier had little chance of saturating on its output. But because modern electronics have moved toward lower supply voltages, the RLD amplifier is more susceptible to saturation; for example, if the RLD amplifier is powered with +3 V, then any attempt to exceed +3 V on its output will cause saturation, at which point the common-mode rejection of the system will immediately drop, and noise can appear on the ECG.

While the RLD system works to cancel out ac common-mode noise on the body, there is another independent system at work in the monitor that is used to detect if the skin–electrode contact has become poor or if the electrodes have fallen off the patient. This so-called dc lead-off system consists of a current source or pull-down resistor attached to the input of each lead, causing a constant dc current to flow through each electrode at all times (see Fig. 1). If an electrode is no longer in solid contact with the skin, this causes the skin–electrode resistance to be increased. This increase will be detected by the dc lead-off system because a dc voltage will appear across the electrode as the skin–electrode resistance increases. If this dc voltage exceeds some threshold, then a message alerts the clinician that an electrode is no longer attached securely to the patient.

One of the unintended side-effects of the dc lead-off system is that it can cause a significant dc voltage to appear on the RLD amplifier output. This is because the sum total of the dc current that is drawn by each of the leads is supplied from the RLD amplifier. For example, if the RA, LA, and LL electrodes are each drawing 100 nA for dc lead-off, this means that the RLD amplifier is sourcing 300 nA. This can cause a dc voltage to appear on the output of the RLD amplifier, particularly when the RL skin–electrode resistance becomes high. For example, if the RL electrode has a resistance of 3 MΩ and there is a 300 nA current flowing through this electrode, then there will be a voltage drop of 0.9 V across the RL electrode (i.e., $V_{\mathrm{RL}} - V_{\mathrm{Body}} = 0.9$ V). Since the RLD system clamps the body voltage very close to ground ($V_{\mathrm{Body}} \sim 0$ V), there will be 0.9 V dc on the output of the RLD amplifier, thereby contributing to saturation.

The possibility that RLD saturation can contribute to ECG noise seen clinically has not been investigated. This has motivated the need for a deeper understanding of what causes RLD saturation and whether or not it can occur in a typical clinical environment. Using a circuit model of an ECG front-end, we explored the relationship between the RLD system, electrode impedance, ac line noise coupling into the body, and dc lead-off current. Our goal was to determine whether saturation could occur under conditions of realistic skin–electrode impedances, and if so, whether this saturation was due primarily to ac or dc on the amplifier output. Understanding the conditions under which the RLD system saturates is critically important for the next generation of ECG monitors that will rely on low-voltage integrated circuits as an analog front-end.

II. Methods

To study RLD saturation, we used the circuit model shown in Fig. 1. Each electrode is modeled as a parallel resistor and capacitor that represents the skin–electrode interface [5], [6]. For example, the RL electrode is made up of a resistance R_{RL}, representing the skin resistance, and capacitor C_{RL}, representing the skin capacitance. The nominal skin–electrode impedance used in the model was defined by the Association for the Advancement of Medical Instrumentation: approximately 50 kΩ for skin resistance and 50 nF for skin capacitance [7]. However, it is well understood that the impedance can vary over time through a number of factors, such as electrode gel drying out, skin moisture level, or loss of secure contact with the skin. For our simulations, we considered the worst case electrode to be 10 MΩ in parallel with 3 nF (see Section IV for the origin of these values). The RA, LA, and LL electrodes, together with the RL electrode, make up the four-electrode test case. Also, to test the commonly used 12-lead ECG (using ten total electrodes), we added six chest leads, each having a skin–electrode impedance of R_P and C_P.

Since the resistance to current flow inside the body is relatively small when compared to the skin–electrode resistance [5], [6], the inside of the body was represented by a single voltage, V_{Body}. We derived our model for the coupling of the mains power to the body from the standards set forth by the International Electrotechnical Commission, including a voltage source of 10 V root-mean-square (rms) [28.3 V peak-to-peak (p-p)] in series with a 200 pF capacitor [8]. This is represented in Fig. 1 by the voltage source, V_{CM}, and capacitor, C_{CM}. Although the mains power is referenced to earth ground, the electronics inside the monitor have a separate signal ground that is isolated from earth ground for safety reasons, as represented by the shaded triangles in Fig. 1. The signal ground within the monitor has some stray capacitance to earth ground, represented by C_g. This capacitance is generally on the order of tens or hundreds of picofarads. For simplicity, we set $C_g = C_{\mathrm{CM}} = 200$ pF. Because signal ground is capacitively coupled to earth ground, any ac voltage on the body with respect to earth will cause current to flow back through the leads, into the monitor, and back to earth ground. Note that this model applies for ECG monitors that are plugged into the mains for power as well as those running on battery power; the only difference between these two cases is that the capacitance C_g will likely be smaller for battery powered systems since the front-end electronics will be more isolated from earth ground, although some finite capacitance will still exist even under battery power.

The RLD system is commonly used in ECG monitors to improve common-mode rejection. The RLD is a feedback system that works by first taking the average of the RA, LA, and LL voltages, referred to as the Wilson voltage ($V_W = [V_{\mathrm{RA}} + V_{\mathrm{LA}} + V_{\mathrm{LL}}]/3$). The Wilson voltage is then applied to

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the inverting input of the RLD amplifier. This amplifier then will drive the voltage V_{RL} to whatever value is necessary to make the Wilson voltage equal to signal ground. The intent of this design is that the RLD amplifier will reduce the amplitude of ac voltages on the body (V_{Body}) with respect to signal ground, thereby greatly improving the common-mode rejection of the overall system. The power supply to the RLD amplifier is represented by $\pm V_{\mathrm{supply}}$; note that the RLD amplifier output voltage cannot exceed the power supply levels ($V_{\mathrm{RL}} < +V_{\mathrm{supply}}$ and $V_{\mathrm{RL}} > -V_{\mathrm{supply}}$), or else saturation will occur. The dc lead-off system consists of several current sources (I_{sense}) that draw a constant dc current from each lead to signal ground. For the simulations, I_{sense} was set to be 100 nA dc, unless otherwise noted.

Simulations were performed in LTSpice. Line frequency is 60 Hz in the United States and 50 Hz in Europe, but we found no appreciable difference in the results when V_{CM} was varied from 50 to 60 Hz. Therefore, results are shown for V_{CM} set to 55 Hz (sinusoidal). The RLD amplifier was configured to have an open loop gain of 10^5, and a gain bandwidth product of 700 kHz. The default solver in LTSpice was used in all cases.

To support our simulation results, we collected data from a custom-made ECG analog front-end based on the circuit diagram shown in Fig. 1. The "patient" was represented by an *RC* network where each electrode was modeled as a parallel resistor and capacitor. To reproduce a full 12-lead ECG, we used ten total electrodes. Line noise was introduced into the "patient" by applying a 55 Hz, 10 V rms sinusoid into body node through a coupling capacitor of 200 pF (as represented by C_{CM} in Fig. 1). This 55 Hz signal was generated by a Wavetek function generator (Model 148A). Note that this signal was generated with respect to earth ground, while all electronics within the ECG front-end were referenced to signal ground that was isolated from earth through an isolation transformer. For our system, the parasitic capacitance C_g was measured to be 160 pF. Current sources were used to generate a sense current of $I_{\mathrm{sense}} = 75$ nA. The power supplies ($\pm V_{\mathrm{supply}}$) were set to ± 2.5 V.

All electrodes were set to 50 kΩ in parallel with 50 nF with two exceptions: 1) the RA electrode was set to zero impedance ($R_{\mathrm{RA}} = 0$ W) in order to introduce some imbalance in electrode impedance on Lead I. This is the standard method used to measure common-mode rejection of an ECG front-end, allowing some of the common-mode signal on the body to be converted to differential noise on the input of the Lead I amplifier, thus reproducing real clinical scenarios in which the electrode impedances will not be perfectly matched. 2) To test the effect of poor RL electrode contact on the RLD system performance, we set C_{RL} to 5 nF and we varied R_{RL} from 0 to 6 MΩ.

We took two series of measurements. First, we measured the dc level on the RLD amplifier output (V_{RL}) using a handheld multimeter (Fluke 179). To avoid loading down the signal, a 1 GΩ resistor was placed in between V_{RL} and the positive input terminal of the meter. The negative terminal was connected to signal ground of the front-end circuitry. Since the input impedance of the scope is 10 MΩ, the signal displayed on the meter is divided down 100:1. The second measurement was that of Lead I ($V_{\mathrm{RA}} - V_{\mathrm{LA}}$), which was obtained using a differential amplifier (see Fig. 1), followed by a bandpass filter with a passband of 0.05–330 Hz (not shown). This signal was then digitized by an analog-to-digital converter with a sampling rate of 2 ksps.

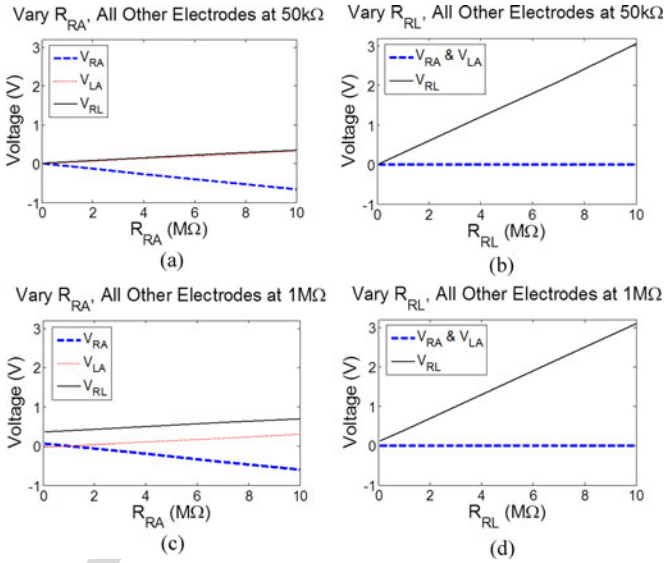


Fig. 2. Examining the interaction between the dc lead-off system and the RLD system. Using a four-electrode ECG with $V_{\mathrm{CM}} = 0$, all electrodes were set to a resistance of 50 kΩ and the resistance of either (a) RA or (b) RL electrode was increased, and the voltage on the RA, RL, and LA is measured. The same test was then repeated for the case where (c, d) all electrodes were initially set to a resistance of 1 MΩ.

III. Results

A. *Effects of DC Lead-Off on RLD Amplifier Saturation*

In our first set of tests, we consider the case where there is no ac input ($V_{\mathrm{CM}} = 0$), allowing us to examine the dc effects of the lead-off current sources on the RLD amplifier output (V_{RL}). All capacitors can be ignored for this test since only dc is considered. Using a four-electrode configuration, we set the resistance of all electrodes to be 50 kΩ. We then varied the resistance of the RA electrode and we see that increases in R_{RA} cause the RA electrode voltage, V_{RA}, to be pulled negative [see Fig. 2(a)]. This is the intended behavior of the lead-off system; a negative voltage on V_{RA} indicates that the RA electrode resistance has increased relative to baseline.

However, there are secondary effects to consider as well. Because V_{RA} has decreased, this has caused a decrease in the Wilson voltage V_W, and any changes in Wilson voltage are counteracted by the RLD system. Specifically, the RLD amplifier will increase V_{RL} in order to clamp the Wilson voltage to ground [see Fig. 2(a)]. Since the dc level of V_{RL} is increased, there is the concern that the RLD amplifier could become saturated on its output. However, in this example, the increase in V_{RL} is quite modest and therefore will have little effect on the saturation of the RLD amplifier. For example, even when R_{RA} reaches 10 MΩ, the RLD output voltage, V_{RL}, is still relatively small (<0.5 V).

Next, the same test was repeated except we looked at the effects of varying the RL electrode resistance R_{RL}. This causes no appreciable change in the RA and LA electrode voltages (V_{RA} and V_{LA}) [see Fig. 2(b)] because the negative feedback

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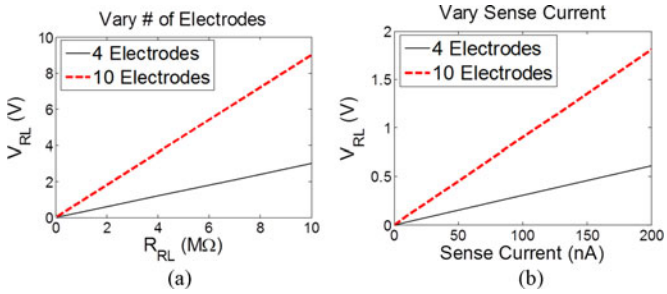


Fig. 3. (a) Effect of varying the number of electrodes from 4 to 10. The RL resistance R_{RL} was increased and the RLD amplifier output voltage V_{RL} was measured. (b) Magnitude of the sense current was varied for the case where $R_{\mathrm{RL}} = 1\ \mathrm{M\Omega}$ and V_{RL} was measured. More electrodes and higher sense current greatly increase the tendency for RLD saturation. For panel (a), I_{sense} was set to 100 nA, while for panel (b), R_{RL} was set to 1 MΩ.

of the RLD system is holding these signals at signal ground potential. However, increasing R_{RL} causes a dramatic increase in the RL electrode voltage V_{RL}. This suggests that increases in R_{RL} could cause the RLD amplifier to become saturated. For example, if $V_{\mathrm{supply}} = 1.5$ V, this would suggest that the RLD amplifier will saturate when the RL resistance is about 5–6 MΩ.

We also considered the case where all of the electrodes had a baseline resistance of 1 MΩ, and then R_{RA} or R_{RL} was increased [see Fig. 2(c) and (d)]. This represents a case where either poor quality electrodes are used or a patient has particularly dry skin. There was a slight vertical shift in the curves when all electrodes were set to 1 MΩ as compared to 50 kΩ [compare Fig. 2(a), (b) to Fig. 2(c), (d)], but the effect was modest, and will not have a significant effect on RLD amplifier saturation.

In the simulations described thus far, we considered ECGs that use four electrodes. Now, we look at the effect of adding six precordial leads, as is used with a full 12-lead ECG (ten electrodes total) [see Fig. 3(a)]. In this case, increasing R_{RL} has a much more pronounced effect on V_{RL} because the current flowing through the RL electrode is the summation of the dc lead-off currents from nine leads ($I_{\mathrm{RL}} = 100\ \mathrm{nA} \times 9 = 900\ \mathrm{nA}$). This means that for the ten-electrode configuration, V_{RL} is now very sensitive to changes in R_{RL}, and as a result, the RLD amplifier may saturate even when the RL electrode is at a relatively low impedance. For example, if the RLD amplifier were powered with $V_{\mathrm{supply}} = 1.5$ V, then the amplifier would saturate when R_{RL} is only 1–2 MΩ. This level of impedance is fairly common in clinical environments (see Section IV), suggesting that low-voltage ECG systems will be susceptible to noisy ECGs when operated with a full 12-leads.

Up to this point, the lead-off sense current has been set to 100 nA, but this value can be varied by the circuit designer. To examine how the magnitude of the sense current affects RLD saturation, we varied I_{sense} from 0 to 200 nA [see Fig. 3(b)]. For this test, R_{RL} was set to 1 MΩ to mimic the effect of a high impedance electrode. There is a clear effect: the larger the sense current, the larger the deviation in V_{RL}, and therefore the more tendency there is for the RLD amplifier to saturate. The obvious way to avoid RLD saturation is to simply use a low enough sense current that saturation never occurs, but unfortunately there are a number of factors in practice that will limit the designer from decreasing the sense current too low (see Section IV).

Since we have removed the dc from our analysis ($V_{\mathrm{CM}} = 0$), we can derive some circuit equations for the system. We are most interested in the RLD amplifier output voltage, and therefore we will express V_{RL} in terms of other circuit parameters. Using Ohm's law on the RL electrode, we have

$$V_{\mathrm{RL}} - V_{\mathrm{Body}} = I_{\mathrm{RL}} \times R_{\mathrm{RL}} \tag{1}$$

where I_{RL} is the current flowing from the RLD amplifier into the body. Since all the electrodes other than RL are drawing current that is equal to I_{sense}, we know that

$$I_{\mathrm{RL}} = (N-1) \times I_{\mathrm{sense}} \tag{2}$$

where *N* is the total number of electrodes. If all electrodes have a relatively low impedance, then we can approximate the body voltage as being equal to the Wilson voltage, and since the Wilson voltage is clamped at 0 V (relative to signal ground), all of the electrode voltages are also clamped at 0 V:

$$V_{\mathrm{Body}} = V_W = V_{\mathrm{RA}} = V_{\mathrm{LA}} = V_{\mathrm{LL}} = 0\,\mathrm{V}. \tag{3}$$

Note that (3) holds only as long as the RLD amplifier is not saturated because it is the RLD system that clamps the Wilson voltage at 0 V. Now, we can express (1) as follows:

$$V_{\mathrm{RL}} = (N-1) \times I_{\mathrm{sense}} \times R_{\mathrm{RL}}. \tag{4}$$

This equation tells us that if the primary electrodes impedances (R_{RA}, R_{LA}, R_{LL}) are much smaller than R_{RL}, then both V_{Body} and V_W are driven to 0 V, and the RLD amplifier output voltage V_{RL} is proportional to the number of electrodes (*N*), the sense current I_{sense}, and the resistance of the RL electrode R_{RL}. This equation describes the slope of the curves in Figs. 2(a), (b) and 3(a), (b).

B. Effect of AC Common-Mode Input on RLD Amplifier Saturation

Thus far, we have shown that the RLD amplifier can saturate as a result of the dc lead-off system. However, the RLD amplifier also puts out an ac signal that will combine with the dc output to further increase the chance of saturation. To determine how much this ac signal will contribute to saturation, we applied a common-mode signal ($V_{\mathrm{CM}} = 10$ V rms) under various electrode conditions.

All of the electrode impedances were set to 50 kΩ in parallel with 50 nF. Two different conditions of RA electrode resistance were tested: $R_{\mathrm{RA}} = 50\ \mathrm{k\Omega}$ and $R_{\mathrm{RA}} = 10\ \mathrm{M\Omega}$. In both cases, V_{RL} has an ac signal on it at 55 Hz [see Fig. 4(a)], and this is expected since the RLD system, by design, is intended to output an ac signal at line frequency to attenuate the common-mode voltage on the body. However, it is interesting that this p-p ac voltage is only 0.1 V, which is much smaller than the 0.35 V dc shift seen when R_{RA} was increased from 50 kΩ to 10 MΩ. We repeated this test for the ten-electrode configuration [see Fig. 4(b)], and in both cases the ac contribution to RLD amplifier saturation was significantly smaller than the dc contribution.

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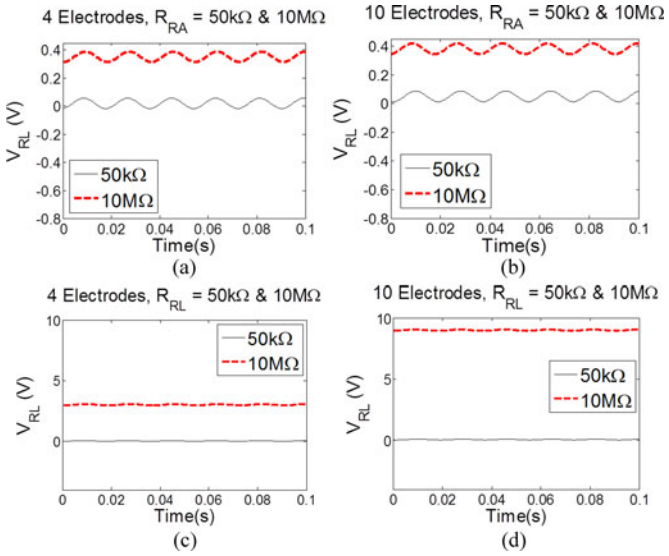


Fig. 4. RLD amplifier output voltage ($V_{\rm RL}$) was measured when an ac common-mode signal was coupled into the body ($V_{\rm CM}$ = 10 V rms). Measurements were made for the case where $R_{\rm RA}$ was set to 50 and 10 MΩ, and this was repeated for (a) four-electrode and (b) ten-electrode configurations. The same test was repeated for the case where (c, d) $R_{\rm RL}$ was varied. Increasing $R_{\rm RA}$ or $R_{\rm RL}$ caused a shift in dc level, but no change in ac amplitude.

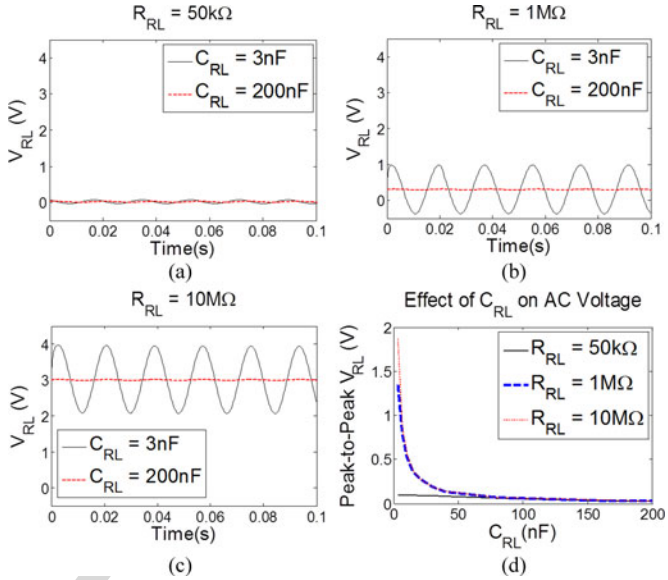


Fig. 5. Effect of varying RL skin capacitance $C_{\rm RL}$ on the ac amplitude of $V_{\rm RL}$ was tested for two cases: good electrode contact (200 nF) and poor electrode contact (3 nF). This test was repeated for across three different conditions of RL electrode resistance $R_{\rm RL}$: (a) 50 kΩ, (b) 1 MΩ, and (c) 10 MΩ. (d) Summary plot was made by measuring the p-p response of $V_{\rm RL}$ for a range of $C_{\rm RL}$ values. All data shown are for the four-electrode configuration.

The reason that the ac voltage dropped across the RL electrode is relatively small can be understood by basic circuit analysis of the model. The current flowing from the voltage source $V_{\rm CM}$ into the monitor flows back to earth ground by passing through two 200 pF capacitors in series, giving a total capacitance of 100 pF. At 55 Hz, a capacitance of 100 pF gives an impedance of –*j*29 MΩ. This current also flows through the RL electrode capacitance, which at 50 nF gives a much lower impedance of –*j*58 kΩ. This means that the 10 V rms is divided between 29 MΩ and 50 kΩ, giving only 17 mV rms (or 48 mV p-p) dropped across the RL electrode capacitance. These calculations are consistent with the simulation results in Fig. 4.

Next, we tested two different conditions of RL electrode resistance: $R_{\rm RL}$ = 50 kΩ and $R_{\rm RL}$ = 10 MΩ, and this was done both for four-electrodes [see Fig. 4(c)] and ten-electrodes [see Fig. 4(d)]. In this case, the ac signal is barely distinguishable when compared to the dramatic shift in dc voltage that is caused by the dc lead-off system [note the difference in scales between Fig. 4(a), (b) versus (c), (d)]. This suggests that varying the resistive component of RL electrode impedance has a dramatic effect on the dc component of the RLD amplifier output, but very little effect on the ac component.

We then tested the effect of the capacitive component of electrode impedance on the ac signal. We considered both the best-case scenario with a large skin–electrode capacitance (200 nF), as well as the worst-case scenario where the skin–electrode contact is poor, giving a low capacitance (3 nF). This was tested for three different values of $R_{\rm RL}$ (50 kΩ, 1 MΩ, 10 MΩ). We see that the ac component of $V_{\rm RL}$ is very small when the electrode resistance is low (50 kΩ) [see Fig. 5(a)]. However, when the electrode resistance is high, the ac signal can become quite large, but only when the capacitance is very low [see Fig. 5(b) and (c)]. In this case, the ac signal may cause the RLD amplifier to saturate. For example, if $V_{\rm supply}$ = 3.5 V, then we can see in Fig. 5(c) that the amplifier will not saturate for the high skin capacitance case, but there will be significant saturation for the low capacitance case as a result of the ac signals.

To summarize how the ac response $V_{\rm RL}$ can affect RLD saturation, we measured the p-p of response over a wide range of skin–electrode capacitances of the RL electrode, $C_{\rm RL}$ [see Fig. 5(d)]. Notice that when the skin–electrode capacitance gets below about 20 nF, there is a dramatic increase in the amplitude of $V_{\rm RL}$, but only when the skin–electrode resistance is relatively high. The reason that $C_{\rm RL}$ has little effect when $R_{\rm RL}$ is low is that total impedance of a parallel *RC* combination will be defined by whichever component is lower in impedance: $R_{\rm RL}$ or $C_{\rm RL}$. For example, at a frequency of 55 Hz, a skin capacitance of 3 nF will impose an impedance of –*j*965 kΩ. Therefore, if the resistance, $R_{\rm RL}$, is «965 kΩ, then the capacitor will be negligible because it is effectively shorted out. It is not until $R_{\rm RL}$ becomes comparable in magnitude to 965 kΩ that the total electrode impedance becomes very high. This, in turn, will cause a significant ac drop across the RL electrode, producing large ac fluctuations in $V_{\rm RL}$.

Finally, we consider how the amplitude of the common-mode signal $V_{\rm CM}$ will influence RLD saturation. The typical model for this common-mode signal consists of a 10 V rms source (28.3 V p-p), but this amplitude can vary across different clinical environments. To account for this, we considered the case where $V_{\rm CM}$ is two times larger than the standard model ($V_{\rm CM}$ = 20 V rms). We simulated four different combinations of $R_{\rm RL}$ and $C_{\rm RL}$, taking into account that increases in skin–electrode resistance tend to be associated with decreases in

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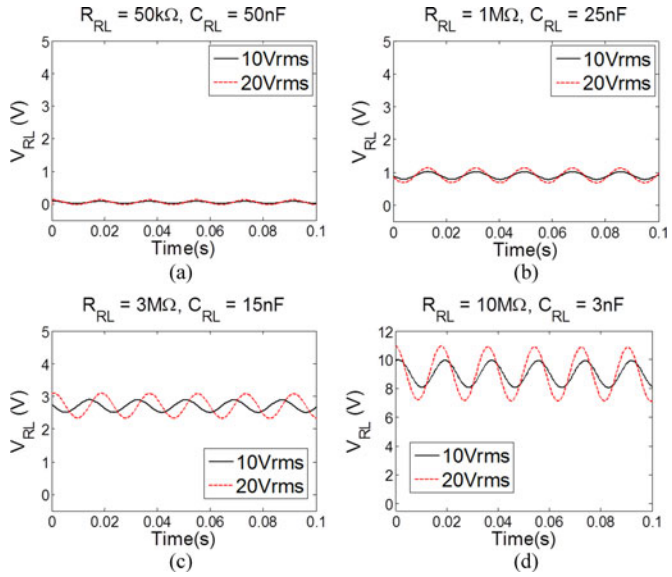


Fig. 6. Effect of increasing common-mode input amplitude was tested for $V_{\mathrm{CM}} = 10$ and 20 V rms using a ten-electrode configuration. The resistance R_{RL} and capacitance C_{RL} of the RL electrode took on four different combinations to mimic real-life conditions in which a poor electrode has increased R_{RL} but decreased C_{RL}. The nominal electrode impedance is (a) 50 kΩ//50 nF, and (b)–(d) electrode becomes increasingly higher in impedance. All other electrodes had an impedance of 50 kΩ//50 nF. Notice the scale is different in panel (d).

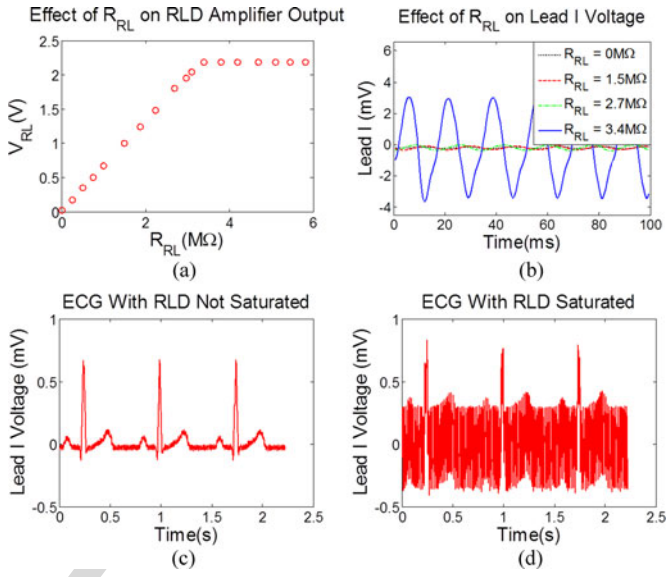


Fig. 7. (a) Effect of R_{RL} on the dc output level of the RLD amplifier was measured. The RLD amplifier saturated when $R_{\mathrm{RL}} = 3.4$ MΩ. (b) Lead I was recorded for a range of RL impedances, illustrating that the saturation of the RLD amplifier causes a sudden increase in ECG noise. (c) and (d) Example of an ECG with and without the RLD amplifier saturated. These noise levels were derived from panel (b) for the cases of (c) $R_{\mathrm{RL}} = 1.5$ MΩ and (d) $R_{\mathrm{RL}} = 3.4$ MΩ. The noise waveforms were scaled down by a factor of 10 to account for any notch filtering at line frequency that might be applied in software.

skin–electrode capacitance. The ten-electrode configuration was used and all electrodes other than RL were set to 50 kΩ//50 nF. Notice that for the cases shown in Fig. 6(a)–(c), the ac magnitude of V_{RL} is very small compared to the dc output, and this is true even when V_{CM} is set to an abnormally high value of 20 V rms. Even when the RL impedance becomes exceptionally poor (10 MΩ in parallel with 3 nF) and the ac amplitude reaches nearly 3 V p-p, it is still the dc component that dominates the V_{RL} output [see Fig. 6(d)]. This suggests that the primary factor that drives RLD amplifier saturation will be dc rather than ac, and this is true even for exceptionally large common-mode inputs (large V_{CM}) with low electrode capacitance (low C_{RL}).

C. *Verifying the Simulation Results With Laboratory Testing*

To illustrate how RLD amplifier saturation can cause a sudden and large increase in noise on the ECG, we conducted a series of experiments using a custom-made ECG analog front-end based on the model shown in Fig. 1 (see Section II). The voltage supply levels were set to ±2.5 V and the sense current amplitude was set to 75 nA; these were chosen to mimic realistic values that modern commercial monitors might use.

In the first set of experiments, we measured the dc voltage on the output of the RLD amplifier (V_{RL}) in response to increases in the dc resistance of the RL electrode R_{RL} [see Fig. 7(a)]. As the model predicted, the RLD output voltage increases in response to increases in RL electrode resistance [compare Figs. 3(a) to 7(a)]. Importantly, the output of the RLD amplifier saturates when the RL impedance becomes sufficiently high. This causes V_{RL} to plateau near the positive power supply rail of the amplifier for $R_{\mathrm{RL}} \geq 3.4$ MΩ. At this point, the RLD amplifier is no longer capable of attenuating the ac common-mode noise on the body, and therefore we hypothesized that the noise on the ECG lead voltages would suddenly increase for $R_{\mathrm{RL}} \geq 3.4$ MΩ.

To test this hypothesis, we conducted a second set of experiments in which Lead I was recorded for different values of R_{RL} [see Fig. 7(b)]. We found that for values of R_{RL} in which the RLD amplifier was not saturated, Lead I has very little signal at 55 Hz, indicating good common-mode rejection of the RLD system. For example, when $R_{\mathrm{RL}} = 1.5$ MΩ, the signal on Lead I has a p-p amplitude of 0.26 mV as compared to the common-mode signal p-p amplitude of $V_{\mathrm{CM}} = 28.3$ V. Therefore, the common-mode rejection ratio of the system is equal to

$$\mathrm{CMRR} = 20 \times \log_{10}\left(\frac{28.3\,\mathrm{V}}{0.26\,\mathrm{mV}}\right) = 100.7\,\mathrm{dB}.$$

Increasing the RL impedance to $R_{\mathrm{RL}} = 2.7$ MΩ caused the noise on Lead I to increase to 0.36 mV p-p, giving a CMRR of 97.9 dB. However, when $R_{\mathrm{RL}} = 3.4$ MΩ, the RLD amplifier is saturated, and the signal on Lead I increased dramatically to 6.7 mV p-p. This results in a CMRR of 72.5 dB. This means that increasing the RL electrode resistance from 2.7 to 3.4 MΩ caused the CMRR to drop by a remarkable 25.4 dB. Further increases in R_{RL} did not cause any further increase in the amplitude of the noise on Lead I (data not shown).

These experimental results confirm what the simulations have suggested that increases in R_{RL} coupled with the use of dc lead-off system can saturate the RLD amplifier and produce significant ECG noise. To illustrate how this noise would appear on a typical patient monitor, we took the noise waveforms displayed

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in Fig 7(b) and added them to a clean ECG waveform [see Fig. 7(c) and (d)] obtained from a patient simulator. In practice, it is common to use a notch filter in software to attenuate any signal at line frequency (50/60 Hz). To account for this, we divided the noise down by a factor of 10 to mimic a 20 dB attenuation by such a filter. The waveforms in Fig. 7(d) illustrate that when the RLD amplifier saturates, line noise will appear as a fat baseline and make the ECG unusable for diagnostic purposes.

IV. Discussion

We developed a circuit model of an ECG front-end to describe the interaction between the RLD system, the dc lead-off system, electrode impedance, and the ac common-mode signal that couples into the body through the power mains. We found that the primary cause of RLD saturation is the dc lead-off system. This is because the RLD amplifier has to source all of the dc current that is pulled by the individual leads, causing a relatively large dc current to flow at all times through the RL electrode. This can cause the RLD amplifier to saturate even under common conditions. For example, with a 12-lead ECG using a 100 nA sense current, we found that saturation would occur when the RL electrode resistance was only 1–2 MΩ, while a four-electrode ECG would saturate at 4–5 MΩ [see Fig. 3(a)].

Given our finding that RLD saturation can occur when the skin–electrode resistance is on the order of 1–5 MΩ, it is important to understand how often such impedances will be observed clinically. Many studies have looked at the range of electrode impedances for various skin types and electrode composition. Unfortunately, these studies often measure the ac impedance using sinusoidal test waveforms while we are interested in the dc resistance of the skin–electrode impedance. But, since the skin capacitance is a relatively high impedance at low frequencies, we can infer the skin–electrode simply by looking at the impedance at low frequencies (1–5 Hz). For example, at 1 Hz, a skin–electrode capacitance of 50 nF has an impedance magnitude of 3.2 MΩ, and therefore the impedance is dominated by the skin–electrode resistance, as long as the resistance is «3.2 MΩ. For example, these studies have found that if the electrodes remain in solid contact with the skin, then the electrode resistance is on the order of tens or hundreds of kilohms, but rarely do they exceed 1 MΩ [9]–[12]. In fact, electrode impedance often *decreases* slightly over time because the electrolyte diffuses conductive ions (Cl^-) into the skin, decreasing skin resistance [12], [13].

However, under real clinical conditions, electrodes usually do not remain in solid contact with the skin, and it is common for electrodes to become dislodged and make only superficial contact with the skin. If the electrode is physically lifted up off of the skin due to pulling of the cables or if there is hair between the electrode and the skin, then the contact area between the gel and the skin becomes very small, thereby increasing the impedance. Under these conditions, the skin–electrode resistance will frequently be greater than 1 MΩ [14]. Furthermore, if an electrode is lifted up off the skin and exposed to air, the gel can dry out and cause the impedance to increase dramatically, reaching over 20 MΩ in some cases [15]. Because it is fairly common for electrodes to be pulled by the cables and make only superficial contact with the skin, we anticipate that skin–electrode resistances of several megohms do occur often in real clinical environments. As a result, we expect that RLD saturation will occur often and this will produce frequent observations of ECG noise in low-voltage monitors.

How can we utilize the information described so far to design an ECG system that uses low-voltage power that will not saturate the RLD amplifier? We offer three possibilities; first, one option is to simply power the RLD amplifier off a higher supply voltage than the rest of the front-end circuitry. This would allow the RLD output voltage to swing as high as necessary during cases of high electrode impedance. This solution is not ideal because having a separate power supply voltage for the RLD system would require more components, and it would necessitate careful design so that all outputs of the RLD amplifier (e.g., switches, buffers) can handle this higher voltage input. Furthermore, while it is possible to have two separate power supply voltages when the front-end is constructed with discrete components, it may not work with ASICs that integrate all of the front-end circuitry.

A second possible design solution to avoid RLD saturation is to simply lower the sense current I_{sense}. For example, we used a nominal sense current of 100 nA, but as Fig. 3(b) shows, reducing sense current will greatly reduce the dc current supplied by the RLD amplifier, thereby reducing the voltage levels on its output. However, there will be a limit on how low the sense current can be reduced. If the current is too low, then other sources of error will appear, such as leakage current across the surface of the printed circuit board, leakage current through any reverse biased diodes that may be present, or input bias current of the amplifiers. This is particularly a problem if pull-down resistors are used instead of real current sources because the current amplitude will decrease as the skin–electrode resistance decreases.

A third possible design solution to avoid RLD saturation is to use an ac-based lead-off system rather than a dc lead-off system. An ac-based method has been proposed in several forms and basically involves the continuous application of some high-frequency ac signal to each electrode. The frequency of this signal is chosen to be beyond the passband of the ECG (0–150 Hz) so as to not interfere with monitoring [16]. While dc lead-off only provides information on the skin–electrode resistance, ac lead-off the benefit of measuring the total skin impedance, including the combination of resistance and capacitance of the skin. However, the challenge with this approach is that the ac stimulation can interfere with other functions of the monitor, such as measuring respiration via impedance pneumography. Likewise, many ECG monitors have the ability to detect and remove pacer artifacts, and this functionality could interfere with an ac lead-off as well.

Each of the solutions described earlier have pros and cons, and the best solution will depend on the particular design specifications of the user. In order to ensure that the RLD system does not saturate under normal operating conditions, it is critically important that the designer understands how the RLD system interacts with the dc lead-off system and the RL electrode resistance. The problem we describe here will become

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even more serious in the coming years as ECG monitors continue to move toward fully integrated ECG front-ends that are lower cost and lower voltage than traditional front-ends built with discrete components.

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Daniel K. Freeman received the B.S. and Ph.D. degrees in biomedical engineering from Boston University, Boston, MA, USA, in 2003 and 2008, respectively, followed by postdoctoral fellowships at MIT and Harvard Medical School.

He performed research for eight years in the areas of retinal electrophysiology and neural stimulation. His current research focus is medical electronics, including analog front-end design for multiparameter patient monitors. He holds the title of Senior Algorithm Engineer with Dräger Medical Systems, Inc., Andover, MA. He has authored numerous articles that have appeared in neuroscience and engineering journals.



Ronald D. Gatzke received the B.S.E.E. degree in electrical engineering from the Case Institute of Technology, Cleveland, OH, USA, and the M.S.E.E. degree from Northeastern University, Boston, MA, USA.

He was with Hewlett-Packard where he was involved in the area of medical electronics, including electro-physiology measurement and ultrasound imaging systems for 32 years. Assignments/titles included Project Engineer, Project Manager, and Engineer Scientist. He then was with Maxim Integrated Products where he was involved in the area of integrated circuits for communications systems, with emphasis on high performance data converters (ADCs and DACs) and high performance RF devices. He held the title of a Senior Scientist. He is currently a Consultant in the area of medical electronics applications. He has authored multiple applications articles for various RF communication journals, presented numerous seminars on ultrasound imaging and cellular communications applications, and authored ten patents.



Georgios Mallas was born in Athens, Greece. He received the B.S. degree in electronics engineering from the Technical Educational Institute of Athens, Athens, in 2006, the M.Sc. degree in electrical engineering from Imperial College, London, U.K., in 2008, and the Ph.D. degree in electrical engineering from Northeastern University, Boston, MA, USA, in 2013.

He is currently with Dräger Medical Systems, Inc., Andover, MA, holding a Senior Algorithm Engineer position. His current research interests include signal processing methods for physiological signals and patient monitoring alarm management.



Yu Chen received the B.S. degree in biomedical and electrical engineering from Xi'an Jiaotong University, Xi'an, China, and the Ph.D. degree in biomedical engineering and instrumentation from Shanghai Jiaotong University, Shanghai, China.

He did biomedical engineering postdoctoral training at the University of Joseph Fourier, Grenoble, France, and University of Minnesota, Minneapolis, MN. He was a Research and Development Manager with Medical Graphics Corporation, St Paul, MN, where he was involved on noninvasive cardiorespiratory diagnostic systems and related software applications. He is currently the Manager of signal acquisition and processing in patient monitoring division of Dräger Medical Systems, Inc., Andover, MA, USA. He has extensive experience of physiological measurement, biomedical signal processing, algorithm development, technology evaluation, intellectual property development, and clinical trials.



Chris J. Brouse was born in Sudbury, ON, Canada. He received the B.S. degree in computer engineering from the University of Ottawa, Ottawa, ON, in 2004, and the M.S. degree in electrical engineering from the University of British Columbia, Vancouver, BC, Canada, in 2007, where he is currently working toward the Ph.D. degree in electrical engineering.

He is currently a Senior Algorithm Engineer at Dräger Medical Systems, Inc., Andover, MA, USA. His specialties include physiological measurement, digital signal processing, and software systems architecture.

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Q1. Author: Please check whether Ref. [3] is OK as amended.

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Saturation of the Right-Leg Drive Amplifier in Low-Voltage ECG Monitors (April 2014)

Daniel K. Freeman*, Ronald D. Gatzke, Georgios Mallas, Yu Chen, and Chris J. Brouse

***Abstract*—Electrocardiogram (ECG) monitoring is a critical tool in patient care, but its utility is often balanced with frustration from clinicians who are constantly distracted by false alarms. This has motivated the need to readdress the major factors that contribute to ECG noise with the goal of reducing false alarms. In this study, we describe a previously unreported phenomenon in which ECG noise can result from an unintended interaction between two systems: 1) the dc lead-off circuitry that is used to detect whether electrodes fall off the patient; and 2) the right-leg drive (RLD) system that is responsible for reducing ac common-mode noise that couples into the body. Using a circuit model to study this interaction, we found that in the presence of a dc lead-off system, even moderate increases in the right-leg skin–electrode resistance can cause the RLD amplifier to saturate. Such saturation can produce ECG noise because the RLD amplifier will no longer be capable of attenuating ac common-mode noise on the body. RLD saturation is particularly a problem for modern ECG monitors that use low-voltage supply levels. For example, for a 12-lead ECG and a 2 V power supply, saturation will occur when the right-leg electrode resistance reaches only 2 MΩ. We discuss several design solutions that can be used in low-voltage monitors to avoid RLD saturation.**

***Index Terms*—Electrocardiogram (ECG), electrode impedance, lead-off, right-leg drive (RLD).**

I. Introduction

ELECTROCARDIOGRAM (ECG) monitors are ubiquitous in modern clinical environments, and their ability to convey reliable, low-noise signals is fundamental to patient monitoring. Because electrical noise can couple into the ECG and interfere with clinical monitoring, there has been significant effort over many decades to understand the origin of such noise [1]. This issue has been given renewed attention due to the fact that ECG noise can trigger false alarms, contributing to the growing problem of "alarm fatigue" [2]. It has been estimated that over 80% of all alarms from patient monitors are false positive, and these excessive alarms can cause clinicians to be distracted from their work and become desensitized to alarms in general, ultimately increasing the number of treatment errors [2], [3], [17]. We hypothesized that ECG monitors being designed today have to contend with a source of ECG noise that was not present even a decade ago. Specifically, there have been recent changes in the semiconductor industry in which the voltage supply levels used to power integrated circuits are much lower than in years past. These lower supply voltages could potentially cause amplifiers in the ECG front-end circuitry to saturate, producing noise on the ECG. To test this hypothesis, we built a circuit model of an analog ECG front-end. We found that, indeed, such saturation can occur as a result of the unintended interaction between two separate circuits: 1) the right-leg drive (RLD) circuitry that suppresses ac common-mode noise on the body [4], [5]; and 2) the dc lead-off system that detects when any electrodes have fallen off of the patient.

To understand how this interaction occurs, it is necessary to first understand how these two systems work in isolation. The RLD system was first introduced into commercial monitors in the 1960s, and it was designed to address the most prominent ECG noise source, that of the mains power (50 Hz in Europe, 60 Hz in the United States). This noise couples into the body through parasitic capacitance to the mains, producing a large ac voltage on the body that is common to all electrodes, hence serving as a source of common-mode noise in the ECG (see Fig. 1). The RLD system works by taking the average voltage appearing on the primary electrodes (right-arm (RA), left-arm (LA),

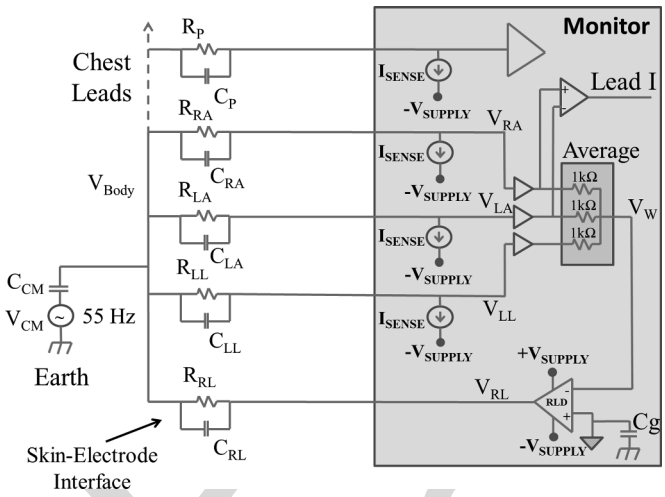


Fig. 1. Illustration of the circuit model that we used to represent a typical ECG system. The body is coupled capacitively to the power mains (V_{CM}) through capacitor C_{CM}. The skin–electrode interface of each electrode is represented by a parallel resistor and capacitor. The RLD system works by averaging the voltages from the primary electrodes (V_{RA}, V_{LA}, V_{LL}) to produce the Wilson voltage (V_W), which is then applied to the inverting input of the RLD amplifier. The dc lead-off system works by pulling current (I_{sense}) through each lead and monitoring the dc voltage on each electrode. The precordial leads are present for the ten-electrode tests but not the four-electrode tests.

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*D. K. Freeman is with the Department of Algorithms, Dräger Medical Systems, Inc., Andover, MA 01810 USA (e-mail: danielkfreeman@gmail.com).

R. D. Gatzke is with Gatzke Technologies, Lexington, MA 02421 USA (e-mail: ron@gatzketech.com).

G. Mallas, Y. Chen, and C. J. Brouse are with the Department of Algorithms, Dräger Medical Systems Inc., Andover, MA 01810 USA (e-mail: Georgios.Mallas@draeger.com; Yu.Chen@draeger.com; Chris.Brouse@draeger.com).

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left-leg (LL)), inverting this signal, and then applying this voltage to the right-leg (RL) electrode to effectively cancel out the ac common-mode noise source on the body [4], [5]. The amplifier that is responsible for driving the inverted signal back into the RL electrode is referred to as the RLD amplifier.

Designers of ECG front-end circuitry must ensure that the RLD amplifier does not saturate on its output during normal operation, or else the RLD system will not function properly. Up until recently, this has not been a major problem because the entire ECG front-end circuitry was typically powered with high voltages (e.g., ±5–10 V), and therefore the RLD amplifier had little chance of saturating on its output. But because modern electronics have moved toward lower supply voltages, the RLD amplifier is more susceptible to saturation; for example, if the RLD amplifier is powered with +3 V, then any attempt to exceed +3 V on its output will cause saturation, at which point the common-mode rejection of the system will immediately drop, and noise can appear on the ECG.

While the RLD system works to cancel out ac common-mode noise on the body, there is another independent system at work in the monitor that is used to detect if the skin–electrode contact has become poor or if the electrodes have fallen off the patient. This so-called dc lead-off system consists of a current source or pull-down resistor attached to the input of each lead, causing a constant dc current to flow through each electrode at all times (see Fig. 1). If an electrode is no longer in solid contact with the skin, this causes the skin–electrode resistance to be increased. This increase will be detected by the dc lead-off system because a dc voltage will appear across the electrode as the skin–electrode resistance increases. If this dc voltage exceeds some threshold, then a message alerts the clinician that an electrode is no longer attached securely to the patient.

One of the unintended side-effects of the dc lead-off system is that it can cause a significant dc voltage to appear on the RLD amplifier output. This is because the sum total of the dc current that is drawn by each of the leads is supplied from the RLD amplifier. For example, if the RA, LA, and LL electrodes are each drawing 100 nA for dc lead-off, this means that the RLD amplifier is sourcing 300 nA. This can cause a dc voltage to appear on the output of the RLD amplifier, particularly when the RL skin–electrode resistance becomes high. For example, if the RL electrode has a resistance of 3 MΩ and there is a 300 nA current flowing through this electrode, then there will be a voltage drop of 0.9 V across the RL electrode (i.e., $V_{\mathrm{RL}} - V_{\mathrm{Body}} = 0.9$ V). Since the RLD system clamps the body voltage very close to ground ($V_{\mathrm{Body}} \sim 0$ V), there will be 0.9 V dc on the output of the RLD amplifier, thereby contributing to saturation.

The possibility that RLD saturation can contribute to ECG noise seen clinically has not been investigated. This has motivated the need for a deeper understanding of what causes RLD saturation and whether or not it can occur in a typical clinical environment. Using a circuit model of an ECG front-end, we explored the relationship between the RLD system, electrode impedance, ac line noise coupling into the body, and dc lead-off current. Our goal was to determine whether saturation could occur under conditions of realistic skin–electrode impedances, and if so, whether this saturation was due primarily to ac or dc on the amplifier output. Understanding the conditions under which the RLD system saturates is critically important for the next generation of ECG monitors that will rely on low-voltage integrated circuits as an analog front-end.

II. Methods

To study RLD saturation, we used the circuit model shown in Fig. 1. Each electrode is modeled as a parallel resistor and capacitor that represents the skin–electrode interface [5], [6]. For example, the RL electrode is made up of a resistance R_{RL}, representing the skin resistance, and capacitor C_{RL}, representing the skin capacitance. The nominal skin–electrode impedance used in the model was defined by the Association for the Advancement of Medical Instrumentation: approximately 50 kΩ for skin resistance and 50 nF for skin capacitance [7]. However, it is well understood that the impedance can vary over time through a number of factors, such as electrode gel drying out, skin moisture level, or loss of secure contact with the skin. For our simulations, we considered the worst case electrode to be 10 MΩ in parallel with 3 nF (see Section IV for the origin of these values). The RA, LA, and LL electrodes, together with the RL electrode, make up the four-electrode test case. Also, to test the commonly used 12-lead ECG (using ten total electrodes), we added six chest leads, each having a skin–electrode impedance of R_P and C_P.

Since the resistance to current flow inside the body is relatively small when compared to the skin–electrode resistance [5], [6], the inside of the body was represented by a single voltage, V_{Body}. We derived our model for the coupling of the mains power to the body from the standards set forth by the International Electrotechnical Commission, including a voltage source of 10 V root-mean-square (rms) [28.3 V peak-to-peak (p-p)] in series with a 200 pF capacitor [8]. This is represented in Fig. 1 by the voltage source, V_{CM}, and capacitor, C_{CM}. Although the mains power is referenced to earth ground, the electronics inside the monitor have a separate signal ground that is isolated from earth ground for safety reasons, as represented by the shaded triangles in Fig. 1. The signal ground within the monitor has some stray capacitance to earth ground, represented by C_g. This capacitance is generally on the order of tens or hundreds of picofarads. For simplicity, we set $C_g = C_{\mathrm{CM}} = 200$ pF. Because signal ground is capacitively coupled to earth ground, any ac voltage on the body with respect to earth will cause current to flow back through the leads, into the monitor, and back to earth ground. Note that this model applies for ECG monitors that are plugged into the mains for power as well as those running on battery power; the only difference between these two cases is that the capacitance C_g will likely be smaller for battery powered systems since the front-end electronics will be more isolated from earth ground, although some finite capacitance will still exist even under battery power.

The RLD system is commonly used in ECG monitors to improve common-mode rejection. The RLD is a feedback system that works by first taking the average of the RA, LA, and LL voltages, referred to as the Wilson voltage ($V_W = [V_{\mathrm{RA}} + V_{\mathrm{LA}} + V_{\mathrm{LL}}]/3$). The Wilson voltage is then applied to

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the inverting input of the RLD amplifier. This amplifier then will drive the voltage V_{RL} to whatever value is necessary to make the Wilson voltage equal to signal ground. The intent of this design is that the RLD amplifier will reduce the amplitude of ac voltages on the body (V_{Body}) with respect to signal ground, thereby greatly improving the common-mode rejection of the overall system. The power supply to the RLD amplifier is represented by $\pm V_{\mathrm{supply}}$; note that the RLD amplifier output voltage cannot exceed the power supply levels ($V_{\mathrm{RL}} < +V_{\mathrm{supply}}$ and $V_{\mathrm{RL}} > -V_{\mathrm{supply}}$), or else saturation will occur. The dc lead-off system consists of several current sources (I_{sense}) that draw a constant dc current from each lead to signal ground. For the simulations, I_{sense} was set to be 100 nA dc, unless otherwise noted.

Simulations were performed in LTSpice. Line frequency is 60 Hz in the United States and 50 Hz in Europe, but we found no appreciable difference in the results when V_{CM} was varied from 50 to 60 Hz. Therefore, results are shown for V_{CM} set to 55 Hz (sinusoidal). The RLD amplifier was configured to have an open loop gain of 10^5, and a gain bandwidth product of 700 kHz. The default solver in LTSpice was used in all cases.

To support our simulation results, we collected data from a custom-made ECG analog front-end based on the circuit diagram shown in Fig. 1. The "patient" was represented by an *RC* network where each electrode was modeled as a parallel resistor and capacitor. To reproduce a full 12-lead ECG, we used ten total electrodes. Line noise was introduced into the "patient" by applying a 55 Hz, 10 V rms sinusoid into body node through a coupling capacitor of 200 pF (as represented by C_{CM} in Fig. 1). This 55 Hz signal was generated by a Wavetek function generator (Model 148A). Note that this signal was generated with respect to earth ground, while all electronics within the ECG front-end were referenced to signal ground that was isolated from earth through an isolation transformer. For our system, the parasitic capacitance C_g was measured to be 160 pF. Current sources were used to generate a sense current of $I_{\mathrm{sense}} = 75$ nA. The power supplies ($\pm V_{\mathrm{supply}}$) were set to $\pm$2.5 V.

All electrodes were set to 50 kΩ in parallel with 50 nF with two exceptions: 1) the RA electrode was set to zero impedance ($R_{\mathrm{RA}} = 0$ W) in order to introduce some imbalance in electrode impedance on Lead I. This is the standard method used to measure common-mode rejection of an ECG front-end, allowing some of the common-mode signal on the body to be converted to differential noise on the input of the Lead I amplifier, thus reproducing real clinical scenarios in which the electrode impedances will not be perfectly matched. 2) To test the effect of poor RL electrode contact on the RLD system performance, we set C_{RL} to 5 nF and we varied R_{RL} from 0 to 6 MΩ.

We took two series of measurements. First, we measured the dc level on the RLD amplifier output (V_{RL}) using a handheld multimeter (Fluke 179). To avoid loading down the signal, a 1 GΩ resistor was placed in between V_{RL} and the positive input terminal of the meter. The negative terminal was connected to signal ground of the front-end circuitry. Since the input impedance of the scope is 10 MΩ, the signal displayed on the meter is divided down 100:1. The second measurement was that of Lead I ($V_{\mathrm{RA}} - V_{\mathrm{LA}}$), which was obtained using a differential amplifier (see Fig. 1), followed by a bandpass filter with a passband of 0.05–330 Hz (not shown). This signal was then digitized by an analog-to-digital converter with a sampling rate of 2 ksps.

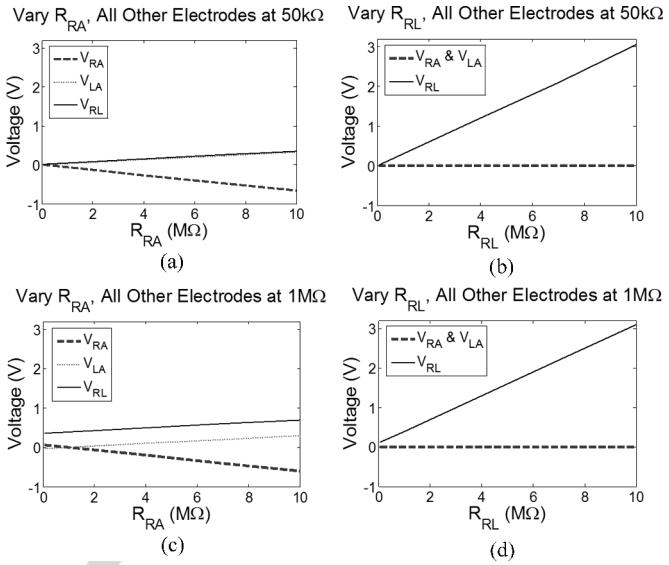


Fig. 2. Examining the interaction between the dc lead-off system and the RLD system. Using a four-electrode ECG with $V_{\mathrm{CM}} = 0$, all electrodes were set to a resistance of 50 kΩ and the resistance of either (a) RA or (b) RL electrode was increased, and the voltage on the RA, RL, and LA is measured. The same test was then repeated for the case where (c, d) all electrodes were initially set to a resistance of 1 MΩ.

III. Results

A. *Effects of DC Lead-Off on RLD Amplifier Saturation*

In our first set of tests, we consider the case where there is no ac input ($V_{\mathrm{CM}} = 0$), allowing us to examine the dc effects of the lead-off current sources on the RLD amplifier output (V_{RL}). All capacitors can be ignored for this test since only dc is considered. Using a four-electrode configuration, we set the resistance of all electrodes to be 50 kΩ. We then varied the resistance of the RA electrode and we see that increases in R_{RA} cause the RA electrode voltage, V_{RA}, to be pulled negative [see Fig. 2(a)]. This is the intended behavior of the lead-off system; a negative voltage on V_{RA} indicates that the RA electrode resistance has increased relative to baseline.

However, there are secondary effects to consider as well. Because V_{RA} has decreased, this has caused a decrease in the Wilson voltage V_W, and any changes in Wilson voltage are counteracted by the RLD system. Specifically, the RLD amplifier will increase V_{RL} in order to clamp the Wilson voltage to ground [see Fig. 2(a)]. Since the dc level of V_{RL} is increased, there is the concern that the RLD amplifier could become saturated on its output. However, in this example, the increase in V_{RL} is quite modest and therefore will have little effect on the saturation of the RLD amplifier. For example, even when R_{RA} reaches 10 MΩ, the RLD output voltage, V_{RL}, is still relatively small (<0.5 V).

Next, the same test was repeated except we looked at the effects of varying the RL electrode resistance R_{RL}. This causes no appreciable change in the RA and LA electrode voltages (V_{RA} and V_{LA}) [see Fig. 2(b)] because the negative feedback

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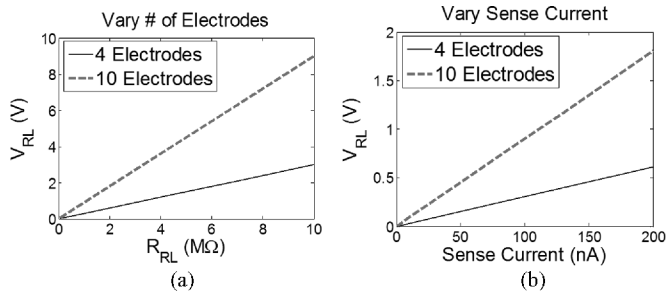


Fig. 3. (a) Effect of varying the number of electrodes from 4 to 10. The RL resistance R_{RL} was increased and the RLD amplifier output voltage V_{RL} was measured. (b) Magnitude of the sense current was varied for the case where $R_{\mathrm{RL}} = 1\,\mathrm{M}\Omega$ and V_{RL} was measured. More electrodes and higher sense current greatly increase the tendency for RLD saturation. For panel (a), I_{sense} was set to 100 nA, while for panel (b), R_{RL} was set to 1 MΩ.

of the RLD system is holding these signals at signal ground potential. However, increasing R_{RL} causes a dramatic increase in the RL electrode voltage V_{RL}. This suggests that increases in R_{RL} could cause the RLD amplifier to become saturated. For example, if $V_{\mathrm{supply}} = 1.5$ V, this would suggest that the RLD amplifier will saturate when the RL resistance is about 5–6 MΩ.

We also considered the case where all of the electrodes had a baseline resistance of 1 MΩ, and then R_{RA} or R_{RL} was increased [see Fig. 2(c) and (d)]. This represents a case where either poor quality electrodes are used or a patient has particularly dry skin. There was a slight vertical shift in the curves when all electrodes were set to 1 MΩ as compared to 50 kΩ [compare Fig. 2(a), (b) to Fig. 2(c), (d)], but the effect was modest, and will not have a significant effect on RLD amplifier saturation.

In the simulations described thus far, we considered ECGs that use four electrodes. Now, we look at the effect of adding six precordial leads, as is used with a full 12-lead ECG (ten electrodes total) [see Fig. 3(a)]. In this case, increasing R_{RL} has a much more pronounced effect on V_{RL} because the current flowing through the RL electrode is the summation of the dc lead-off currents from nine leads ($I_{\mathrm{RL}} = 100\,\mathrm{nA} \times 9 = 900\,\mathrm{nA}$). This means that for the ten-electrode configuration, V_{RL} is now very sensitive to changes in R_{RL}, and as a result, the RLD amplifier may saturate even when the RL electrode is at a relatively low impedance. For example, if the RLD amplifier were powered with $V_{\mathrm{supply}} = 1.5$ V, then the amplifier would saturate when R_{RL} is only 1–2 MΩ. This level of impedance is fairly common in clinical environments (see Section IV), suggesting that low-voltage ECG systems will be susceptible to noisy ECGs when operated with a full 12-leads.

Up to this point, the lead-off sense current has been set to 100 nA, but this value can be varied by the circuit designer. To examine how the magnitude of the sense current affects RLD saturation, we varied I_{sense} from 0 to 200 nA [see Fig. 3(b)]. For this test, R_{RL} was set to 1 MΩ to mimic the effect of a high impedance electrode. There is a clear effect: the larger the sense current, the larger the deviation in V_{RL}, and therefore the more tendency there is for the RLD amplifier to saturate. The obvious way to avoid RLD saturation is to simply use a low enough sense current that saturation never occurs, but unfortunately there are a number of factors in practice that will limit the designer from decreasing the sense current too low (see Section IV).

Since we have removed the dc from our analysis ($V_{\mathrm{CM}} = 0$), we can derive some circuit equations for the system. We are most interested in the RLD amplifier output voltage, and therefore we will express V_{RL} in terms of other circuit parameters. Using Ohm's law on the RL electrode, we have

$$V_{\mathrm{RL}} - V_{\mathrm{Body}} = I_{\mathrm{RL}} \times R_{\mathrm{RL}} \tag{1}$$

where I_{RL} is the current flowing from the RLD amplifier into the body. Since all the electrodes other than RL are drawing current that is equal to I_{sense}, we know that

$$I_{\mathrm{RL}} = (N-1) \times I_{\mathrm{sense}} \tag{2}$$

where *N* is the total number of electrodes. If all electrodes have a relatively low impedance, then we can approximate the body voltage as being equal to the Wilson voltage, and since the Wilson voltage is clamped at 0 V (relative to signal ground), all of the electrode voltages are also clamped at 0 V:

$$V_{\mathrm{Body}} = V_W = V_{\mathrm{RA}} = V_{\mathrm{LA}} = V_{\mathrm{LL}} = 0\,\mathrm{V}. \tag{3}$$

Note that (3) holds only as long as the RLD amplifier is not saturated because it is the RLD system that clamps the Wilson voltage at 0 V. Now, we can express (1) as follows:

$$V_{\mathrm{RL}} = (N-1) \times I_{\mathrm{sense}} \times R_{\mathrm{RL}}. \tag{4}$$

This equation tells us that if the primary electrodes impedances (R_{RA}, R_{LA}, R_{LL}) are much smaller than R_{RL}, then both V_{Body} and V_W are driven to 0 V, and the RLD amplifier output voltage V_{RL} is proportional to the number of electrodes (*N*), the sense current I_{sense}, and the resistance of the RL electrode R_{RL}. This equation describes the slope of the curves in Figs. 2(a), (b) and 3(a), (b).

B. Effect of AC Common-Mode Input on RLD Amplifier Saturation

Thus far, we have shown that the RLD amplifier can saturate as a result of the dc lead-off system. However, the RLD amplifier also puts out an ac signal that will combine with the dc output to further increase the chance of saturation. To determine how much this ac signal will contribute to saturation, we applied a common-mode signal ($V_{\mathrm{CM}} = 10$ V rms) under various electrode conditions.

All of the electrode impedances were set to 50 kΩ in parallel with 50 nF. Two different conditions of RA electrode resistance were tested: $R_{\mathrm{RA}} = 50\,\mathrm{k}\Omega$ and $R_{\mathrm{RA}} = 10\,\mathrm{M}\Omega$. In both cases, V_{RL} has an ac signal on it at 55 Hz [see Fig. 4(a)], and this is expected since the RLD system, by design, is intended to output an ac signal at line frequency to attenuate the common-mode voltage on the body. However, it is interesting that this p-p ac voltage is only 0.1 V, which is much smaller than the 0.35 V dc shift seen when R_{RA} was increased from 50 kΩ to 10 MΩ. We repeated this test for the ten-electrode configuration [see Fig. 4(b)], and in both cases the ac contribution to RLD amplifier saturation was significantly smaller than the dc contribution.

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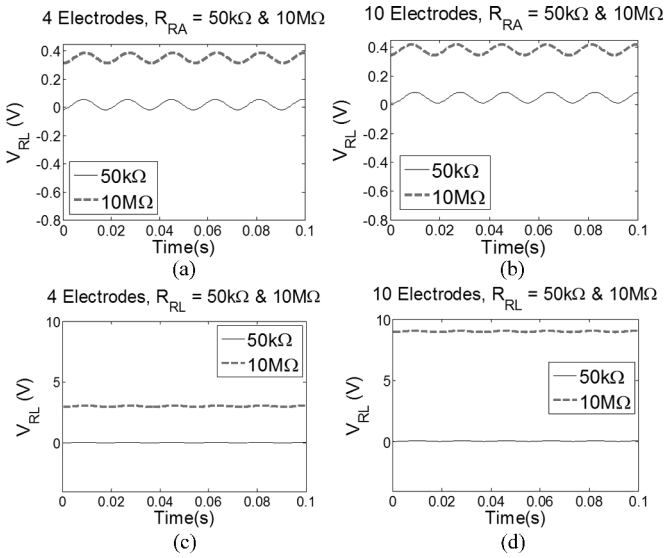


Fig. 4. RLD amplifier output voltage ($V_{\rm RL}$) was measured when an ac common-mode signal was coupled into the body ($V_{\rm CM}$ = 10 V rms). Measurements were made for the case where $R_{\rm RA}$ was set to 50 and 10 MΩ, and this was repeated for (a) four-electrode and (b) ten-electrode configurations. The same test was repeated for the case where (c, d) $R_{\rm RL}$ was varied. Increasing $R_{\rm RA}$ or $R_{\rm RL}$ caused a shift in dc level, but no change in ac amplitude.

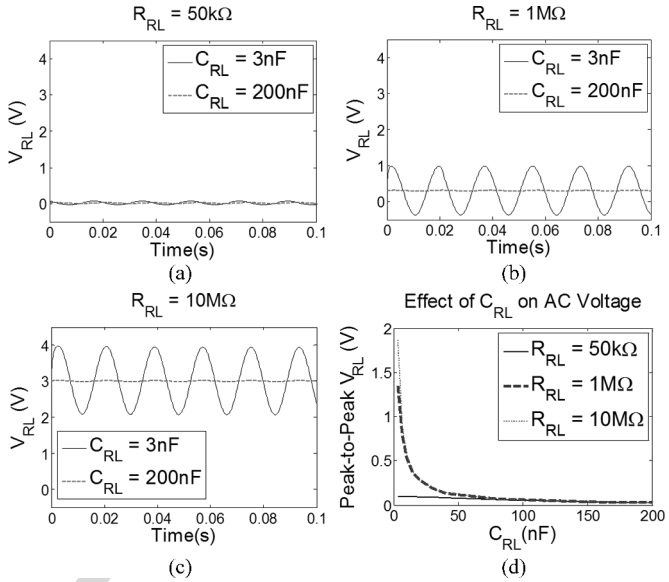


Fig. 5. Effect of varying RL skin capacitance $C_{\rm RL}$ on the ac amplitude of $V_{\rm RL}$ was tested for two cases: good electrode contact (200 nF) and poor electrode contact (3 nF). This test was repeated for across three different conditions of RL electrode resistance $R_{\rm RL}$: (a) 50 kΩ, (b) 1 MΩ, and (c) 10 MΩ. (d) Summary plot was made by measuring the p-p response of $V_{\rm RL}$ for a range of $C_{\rm RL}$ values. All data shown are for the four-electrode configuration.

The reason that the ac voltage dropped across the RL electrode is relatively small can be understood by basic circuit analysis of the model. The current flowing from the voltage source $V_{\rm CM}$ into the monitor flows back to earth ground by passing through two 200 pF capacitors in series, giving a total capacitance of 100 pF. At 55 Hz, a capacitance of 100 pF gives an impedance of –*j*29 MΩ. This current also flows through the RL electrode capacitance, which at 50 nF gives a much lower impedance of –*j*58 kΩ. This means that the 10 V rms is divided between 29 MΩ and 50 kΩ, giving only 17 mV rms (or 48 mV p-p) dropped across the RL electrode capacitance. These calculations are consistent with the simulation results in Fig. 4.

Next, we tested two different conditions of RL electrode resistance: $R_{\rm RL}$ = 50 kΩ and $R_{\rm RL}$ = 10 MΩ, and this was done both for four-electrodes [see Fig. 4(c)] and ten-electrodes [see Fig. 4(d)]. In this case, the ac signal is barely distinguishable when compared to the dramatic shift in dc voltage that is caused by the dc lead-off system [note the difference in scales between Fig. 4(a), (b) versus (c), (d)]. This suggests that varying the resistive component of RL electrode impedance has a dramatic effect on the dc component of the RLD amplifier output, but very little effect on the ac component.

We then tested the effect of the capacitive component of electrode impedance on the ac signal. We considered both the best-case scenario with a large skin–electrode capacitance (200 nF), as well as the worst-case scenario where the skin–electrode contact is poor, giving a low capacitance (3 nF). This was tested for three different values of $R_{\rm RL}$ (50 kΩ, 1 MΩ, 10 MΩ). We see that the ac component of $V_{\rm RL}$ is very small when the electrode resistance is low (50 kΩ) [see Fig. 5(a)]. However, when the electrode resistance is high, the ac signal can become quite large, but only when the capacitance is very low [see Fig. 5(b) and (c)]. In this case, the ac signal may cause the RLD amplifier to saturate. For example, if $V_{\rm supply}$ = 3.5 V, then we can see in Fig. 5(c) that the amplifier will not saturate for the high skin capacitance case, but there will be significant saturation for the low capacitance case as a result of the ac signals.

To summarize how the ac response $V_{\rm RL}$ can affect RLD saturation, we measured the p-p of response over a wide range of skin–electrode capacitances of the RL electrode, $C_{\rm RL}$ [see Fig. 5(d)]. Notice that when the skin–electrode capacitance gets below about 20 nF, there is a dramatic increase in the amplitude of $V_{\rm RL}$, but only when the skin–electrode resistance is relatively high. The reason that $C_{\rm RL}$ has little effect when $R_{\rm RL}$ is low is that total impedance of a parallel *RC* combination will be defined by whichever component is lower in impedance: $R_{\rm RL}$ or $C_{\rm RL}$. For example, at a frequency of 55 Hz, a skin capacitance of 3 nF will impose an impedance of –*j*965 kΩ. Therefore, if the resistance, $R_{\rm RL}$, is «965 kΩ, then the capacitor will be negligible because it is effectively shorted out. It is not until $R_{\rm RL}$ becomes comparable in magnitude to 965 kΩ that the total electrode impedance becomes very high. This, in turn, will cause a significant ac drop across the RL electrode, producing large ac fluctuations in $V_{\rm RL}$.

Finally, we consider how the amplitude of the common-mode signal $V_{\rm CM}$ will influence RLD saturation. The typical model for this common-mode signal consists of a 10 V rms source (28.3 V p-p), but this amplitude can vary across different clinical environments. To account for this, we considered the case where $V_{\rm CM}$ is two times larger than the standard model ($V_{\rm CM}$ = 20 V rms). We simulated four different combinations of $R_{\rm RL}$ and $C_{\rm RL}$, taking into account that increases in skin–electrode resistance tend to be associated with decreases in

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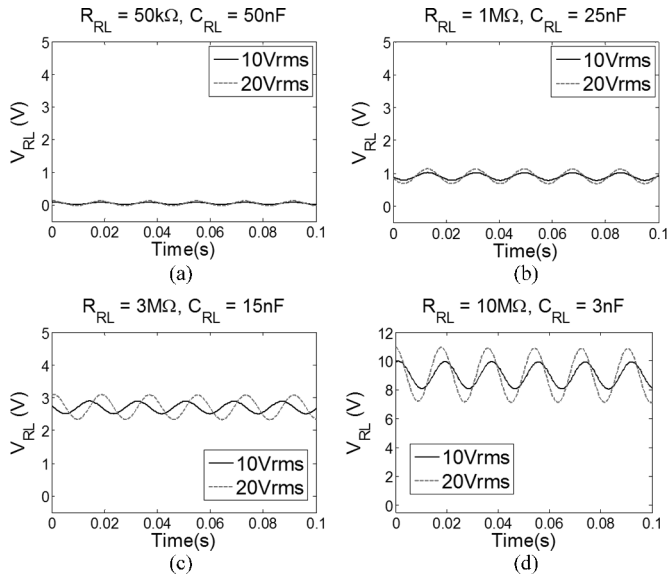


Fig. 6. Effect of increasing common-mode input amplitude was tested for $V_{\mathrm{CM}} = 10$ and 20 V rms using a ten-electrode configuration. The resistance R_{RL} and capacitance C_{RL} of the RL electrode took on four different combinations to mimic real-life conditions in which a poor electrode has increased R_{RL} but decreased C_{RL}. The nominal electrode impedance is (a) 50 kΩ//50 nF, and (b)–(d) electrode becomes increasingly higher in impedance. All other electrodes had an impedance of 50 kΩ//50 nF. Notice the scale is different in panel (d).

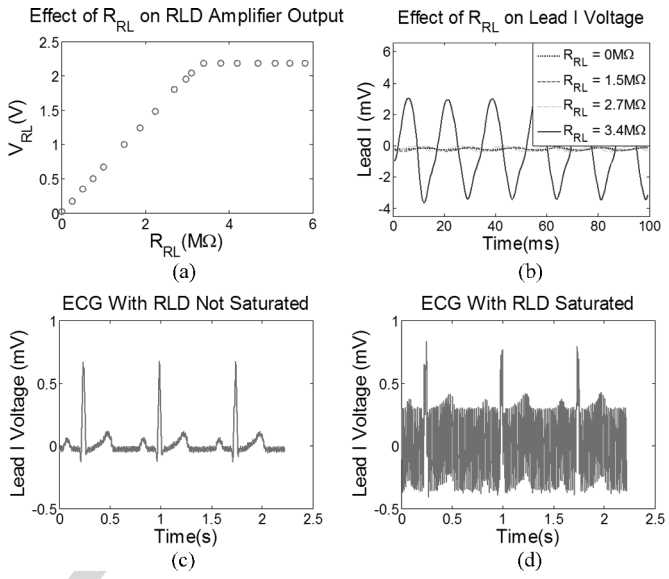


Fig. 7. (a) Effect of R_{RL} on the dc output level of the RLD amplifier was measured. The RLD amplifier saturated when $R_{\mathrm{RL}} = 3.4$ MΩ. (b) Lead I was recorded for a range of RL impedances, illustrating that the saturation of the RLD amplifier causes a sudden increase in ECG noise. (c) and (d) Example of an ECG with and without the RLD amplifier saturated. These noise levels were derived from panel (b) for the cases of (c) $R_{\mathrm{RL}} = 1.5$ MΩ and (d) $R_{\mathrm{RL}} = 3.4$ MΩ. The noise waveforms were scaled down by a factor of 10 to account for any notch filtering at line frequency that might be applied in software.

skin–electrode capacitance. The ten-electrode configuration was used and all electrodes other than RL were set to 50 kΩ//50 nF. Notice that for the cases shown in Fig. 6(a)–(c), the ac magnitude of V_{RL} is very small compared to the dc output, and this is true even when V_{CM} is set to an abnormally high value of 20 V rms. Even when the RL impedance becomes exceptionally poor (10 MΩ in parallel with 3 nF) and the ac amplitude reaches nearly 3 V p-p, it is still the dc component that dominates the V_{RL} output [see Fig. 6(d)]. This suggests that the primary factor that drives RLD amplifier saturation will be dc rather than ac, and this is true even for exceptionally large common-mode inputs (large V_{CM}) with low electrode capacitance (low C_{RL}).

C. Verifying the Simulation Results With Laboratory Testing

To illustrate how RLD amplifier saturation can cause a sudden and large increase in noise on the ECG, we conducted a series of experiments using a custom-made ECG analog front-end based on the model shown in Fig. 1 (see Section II). The voltage supply levels were set to ±2.5 V and the sense current amplitude was set to 75 nA; these were chosen to mimic realistic values that modern commercial monitors might use.

In the first set of experiments, we measured the dc voltage on the output of the RLD amplifier (V_{RL}) in response to increases in the dc resistance of the RL electrode R_{RL} [see Fig. 7(a)]. As the model predicted, the RLD output voltage increases in response to increases in RL electrode resistance [compare Figs. 3(a) to 7(a)]. Importantly, the output of the RLD amplifier saturates when the RL impedance becomes sufficiently high. This causes V_{RL} to plateau near the positive power supply rail of the amplifier for $R_{\mathrm{RL}} \geq 3.4$ MΩ. At this point, the RLD amplifier is no longer capable of attenuating the ac common-mode noise on the body, and therefore we hypothesized that the noise on the ECG lead voltages would suddenly increase for $R_{\mathrm{RL}} \geq 3.4$ MΩ.

To test this hypothesis, we conducted a second set of experiments in which Lead I was recorded for different values of R_{RL} [see Fig. 7(b)]. We found that for values of R_{RL} in which the RLD amplifier was not saturated, Lead I has very little signal at 55 Hz, indicating good common-mode rejection of the RLD system. For example, when $R_{\mathrm{RL}} = 1.5$ MΩ, the signal on Lead I has a p-p amplitude of 0.26 mV as compared to the common-mode signal p-p amplitude of $V_{\mathrm{CM}} = 28.3$ V. Therefore, the common-mode rejection ratio of the system is equal to

$$\mathrm{CMRR} = 20 \times \log_{10}\left(\frac{28.3\,\mathrm{V}}{0.26\,\mathrm{mV}}\right) = 100.7\,\mathrm{dB}.$$

Increasing the RL impedance to $R_{\mathrm{RL}} = 2.7$ MΩ caused the noise on Lead I to increase to 0.36 mV p-p, giving a CMRR of 97.9 dB. However, when $R_{\mathrm{RL}} = 3.4$ MΩ, the RLD amplifier is saturated, and the signal on Lead I increased dramatically to 6.7 mV p-p. This results in a CMRR of 72.5 dB. This means that increasing the RL electrode resistance from 2.7 to 3.4 MΩ caused the CMRR to drop by a remarkable 25.4 dB. Further increases in R_{RL} did not cause any further increase in the amplitude of the noise on Lead I (data not shown).

These experimental results confirm what the simulations have suggested that increases in R_{RL} coupled with the use of dc lead-off system can saturate the RLD amplifier and produce significant ECG noise. To illustrate how this noise would appear on a typical patient monitor, we took the noise waveforms displayed

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in Fig 7(b) and added them to a clean ECG waveform [see Fig. 7(c) and (d)] obtained from a patient simulator. In practice, it is common to use a notch filter in software to attenuate any signal at line frequency (50/60 Hz). To account for this, we divided the noise down by a factor of 10 to mimic a 20 dB attenuation by such a filter. The waveforms in Fig. 7(d) illustrate that when the RLD amplifier saturates, line noise will appear as a fat baseline and make the ECG unusable for diagnostic purposes.

IV. Discussion

We developed a circuit model of an ECG front-end to describe the interaction between the RLD system, the dc lead-off system, electrode impedance, and the ac common-mode signal that couples into the body through the power mains. We found that the primary cause of RLD saturation is the dc lead-off system. This is because the RLD amplifier has to source all of the dc current that is pulled by the individual leads, causing a relatively large dc current to flow at all times through the RL electrode. This can cause the RLD amplifier to saturate even under common conditions. For example, with a 12-lead ECG using a 100 nA sense current, we found that saturation would occur when the RL electrode resistance was only 1–2 MΩ, while a four-electrode ECG would saturate at 4–5 MΩ [see Fig. 3(a)].

Given our finding that RLD saturation can occur when the skin–electrode resistance is on the order of 1–5 MΩ, it is important to understand how often such impedances will be observed clinically. Many studies have looked at the range of electrode impedances for various skin types and electrode composition. Unfortunately, these studies often measure the ac impedance using sinusoidal test waveforms while we are interested in the dc resistance of the skin–electrode impedance. But, since the skin capacitance is a relatively high impedance at low frequencies, we can infer the skin–electrode simply by looking at the impedance at low frequencies (1–5 Hz). For example, at 1 Hz, a skin–electrode capacitance of 50 nF has an impedance magnitude of 3.2 MΩ, and therefore the impedance is dominated by the skin–electrode resistance, as long as the resistance is «3.2 MΩ. For example, these studies have found that if the electrodes remain in solid contact with the skin, then the electrode resistance is on the order of tens or hundreds of kilohms, but rarely do they exceed 1 MΩ [9]–[12]. In fact, electrode impedance often *decreases* slightly over time because the electrolyte diffuses conductive ions (Cl^-) into the skin, decreasing skin resistance [12], [13].

However, under real clinical conditions, electrodes usually do not remain in solid contact with the skin, and it is common for electrodes to become dislodged and make only superficial contact with the skin. If the electrode is physically lifted up off of the skin due to pulling of the cables or if there is hair between the electrode and the skin, then the contact area between the gel and the skin becomes very small, thereby increasing the impedance. Under these conditions, the skin–electrode resistance will frequently be greater than 1 MΩ [14]. Furthermore, if an electrode is lifted up off the skin and exposed to air, the gel can dry out and cause the impedance to increase dramatically, reaching over 20 MΩ in some cases [15]. Because it is fairly common for electrodes to be pulled by the cables and make only superficial contact with the skin, we anticipate that skin–electrode resistances of several megohms do occur often in real clinical environments. As a result, we expect that RLD saturation will occur often and this will produce frequent observations of ECG noise in low-voltage monitors.

How can we utilize the information described so far to design an ECG system that uses low-voltage power that will not saturate the RLD amplifier? We offer three possibilities; first, one option is to simply power the RLD amplifier off a higher supply voltage than the rest of the front-end circuitry. This would allow the RLD output voltage to swing as high as necessary during cases of high electrode impedance. This solution is not ideal because having a separate power supply voltage for the RLD system would require more components, and it would necessitate careful design so that all outputs of the RLD amplifier (e.g., switches, buffers) can handle this higher voltage input. Furthermore, while it is possible to have two separate power supply voltages when the front-end is constructed with discrete components, it may not work with ASICs that integrate all of the front-end circuitry.

A second possible design solution to avoid RLD saturation is to simply lower the sense current I_{sense}. For example, we used a nominal sense current of 100 nA, but as Fig. 3(b) shows, reducing sense current will greatly reduce the dc current supplied by the RLD amplifier, thereby reducing the voltage levels on its output. However, there will be a limit on how low the sense current can be reduced. If the current is too low, then other sources of error will appear, such as leakage current across the surface of the printed circuit board, leakage current through any reverse biased diodes that may be present, or input bias current of the amplifiers. This is particularly a problem if pull-down resistors are used instead of real current sources because the current amplitude will decrease as the skin–electrode resistance decreases.

A third possible design solution to avoid RLD saturation is to use an ac-based lead-off system rather than a dc lead-off system. An ac-based method has been proposed in several forms and basically involves the continuous application of some high-frequency ac signal to each electrode. The frequency of this signal is chosen to be beyond the passband of the ECG (0–150 Hz) so as to not interfere with monitoring [16]. While dc lead-off only provides information on the skin–electrode resistance, ac lead-off the benefit of measuring the total skin impedance, including the combination of resistance and capacitance of the skin. However, the challenge with this approach is that the ac stimulation can interfere with other functions of the monitor, such as measuring respiration via impedance pneumography. Likewise, many ECG monitors have the ability to detect and remove pacer artifacts, and this functionality could interfere with an ac lead-off as well.

Each of the solutions described earlier have pros and cons, and the best solution will depend on the particular design specifications of the user. In order to ensure that the RLD system does not saturate under normal operating conditions, it is critically important that the designer understands how the RLD system interacts with the dc lead-off system and the RL electrode resistance. The problem we describe here will become

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even more serious in the coming years as ECG monitors continue to move toward fully integrated ECG front-ends that are lower cost and lower voltage than traditional front-ends built with discrete components.

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Daniel K. Freeman received the B.S. and Ph.D. degrees in biomedical engineering from Boston University, Boston, MA, USA, in 2003 and 2008, respectively, followed by postdoctoral fellowships at MIT and Harvard Medical School.

He performed research for eight years in the areas of retinal electrophysiology and neural stimulation. His current research focus is medical electronics, including analog front-end design for multiparameter patient monitors. He holds the title of Senior Algorithm Engineer with Dräger Medical Systems, Inc., Andover, MA. He has authored numerous articles that have appeared in neuroscience and engineering journals.



Ronald D. Gatzke received the B.S.E.E. degree in electrical engineering from the Case Institute of Technology, Cleveland, OH, USA, and the M.S.E.E. degree from Northeastern University, Boston, MA, USA.

He was with Hewlett-Packard where he was involved in the area of medical electronics, including electro-physiology measurement and ultrasound imaging systems for 32 years. Assignments/titles included Project Engineer, Project Manager, and Engineer Scientist. He then was with Maxim Integrated Products where he was involved in the area of integrated circuits for communications systems, with emphasis on high performance data converters (ADCs and DACs) and high performance RF devices. He held the title of a Senior Scientist. He is currently a Consultant in the area of medical electronics applications. He has authored multiple applications articles for various RF communication journals, presented numerous seminars on ultrasound imaging and cellular communications applications, and authored ten patents.



Georgios Mallas was born in Athens, Greece. He received the B.S. degree in electronics engineering from the Technical Educational Institute of Athens, Athens, in 2006, the M.Sc. degree in electrical engineering from Imperial College, London, U.K., in 2008, and the Ph.D. degree in electrical engineering from Northeastern University, Boston, MA, USA, in 2013.

He is currently with Dräger Medical Systems, Inc., Andover, MA, holding a Senior Algorithm Engineer position. His current research interests include signal processing methods for physiological signals and patient monitoring alarm management.



Yu Chen received the B.S. degree in biomedical and electrical engineering from Xi'an Jiaotong University, Xi'an, China, and the Ph.D. degree in biomedical engineering and instrumentation from Shanghai Jiaotong University, Shanghai, China.

He did biomedical engineering postdoctoral training at the University of Joseph Fourier, Grenoble, France, and University of Minnesota, Minneapolis, MN. He was a Research and Development Manager with Medical Graphics Corporation, St Paul, MN, where he was involved on noninvasive cardiorespiratory diagnostic systems and related software applications. He is currently the Manager of signal acquisition and processing in patient monitoring division of Dräger Medical Systems, Inc., Andover, MA, USA. He has extensive experience of physiological measurement, biomedical signal processing, algorithm development, technology evaluation, intellectual property development, and clinical trials.



Chris J. Brouse was born in Sudbury, ON, Canada. He received the B.S. degree in computer engineering from the University of Ottawa, Ottawa, ON, in 2004, and the M.S. degree in electrical engineering from the University of British Columbia, Vancouver, BC, Canada, in 2007, where he is currently working toward the Ph.D. degree in electrical engineering.

He is currently a Senior Algorithm Engineer at Dräger Medical Systems, Inc., Andover, MA, USA. His specialties include physiological measurement, digital signal processing, and software systems architecture.

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Q1. Author: Please check whether Ref. [3] is OK as amended.

Q2. Author: Ref. [3] has been split into two references, i.e., [3] and [17], and accordingly they have been cited in the text. Please check.

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